Real Estate Finance App — A-to-Z Feature & Logic Spec

Source: Shopno Bilash Properties admin panel (Next.js + Prisma). Purpose: give the Finance App everything the admin panel already does, so nothing is missed. Rules marked **MUST** are non-negotiable because wrong numbers = real money loss.

1. Big picture

The admin panel is a real-estate developer platform. Money flows in 3 directions, and the Finance App must cover all of them:

Flow	Who pays whom	Examples
Money IN (sales)	Customers → Company	Booking money, down payment, installments, handover payments
Money IN (funding)	Owners / Shareholders / Landowners → Company	Cost-allocation contributions per construction goal, with fines on delay
Money OUT	Company → Contractors, Vendors, Staff, Govt.	Material purchases, contractor payments, project expenses, office/staff costs

Hierarchy: `Project → Building → Floor → Unit`. Every finance record carries an optional `projectId` so reports can be scoped per project or company-wide.

2. Roles & permissions

Role	Finance access
Super Admin	Everything incl. roles/permissions
Managing Director / Owner	Everything except roles/permissions management
Finance Manager	All finance + reports; read-only on project/unit/customer
Project / Construction Manager	Project, unit, construction; cost visibility per project
Sales Manager / Executive	Leads, bookings, sales, customers; payments they record
Procurement Manager	Materials, purchases, stock
Document/Legal Manager	Documents only
Customer / Shareholder / Landowner	Portal only: see *own* units, payments, contributions, agreements

Every create/edit/delete/login writes an append-only **audit log** (who, what, entity, time, IP). Finance App MUST keep this.

3. Modules & features (complete list)

3.1 Projects, Buildings, Floors, Units

- Project: name, location, status, dates; sub-pages: Units, Parking, Construction, Documents, Expenses, Materials, Owners, Payments, Reports, Settings.
- Building → Floor → Unit (flat/shop). Unit: size, price, status (available / booked / sold / owner-allocated / landowner-share).
- A unit belongs to **exactly one** of: Customer (sold/booked), Shareholder allocation, Landowner allocation — never two at once.
- Parking spaces (types, price, ownership assignment) and standalone Property listings (public website).

3.2 CRM / Leads

- Lead: name, phone, interested project/unit, source, assigned salesperson, status; activity log (calls, visits, notes).
- Lead → Customer conversion; reminders and notifications for follow-ups.

3.3 Customers

- Profile, NID/contact, linked units, documents, payment history, outstanding balance, portal login (phone-based).

3.4 Sales workflow

1. **Booking** — reserve a unit with booking money (unit becomes "booked").
2. **Sale / Contract** — agreed price, discounts, payment plan; unit becomes "sold".
3. **Installments** — schedule of due dates and amounts per contract.
4. **Payments** — each receipt: amount, date, method, reference, receipt; allocated against installments.
5. **Cancellation** — cancelled bookings/sales are excluded from totals; unit returns to available.

Sales reports	What it shows
Sales dashboard	Customers, sales count, total sales value, leads/enquiries
Receivables	Per customer: contract value, paid, outstanding, overdue installments
Collection report	Payments received by period / project / method

3.5 Shareholders & Landowners

- Shareholder → Shareholdings per project (percentage/units) → contributions and payouts.
- Landowner → Agreements (e.g. developer–landowner ratio, e.g. 40:60) → Allocations of specific units, plus any cash/advance terms.

3.6 Cost Allocation / Owner Contributions ("Construction Goals")

When construction needs funding, admin creates a **Cost Allocation (Goal)** with total amount, date, fine rule and installments, then the system splits it across owners (by share/unit). Each owner gets an **Owner Contribution** (payable amount, due date, paid amount) and optional per-installment obligations. Admin can add **Adjustments** (waiver / discount / correction, ± amount) — the original payable is never changed.

3.7 Procurement & Stock

- Vendors, Material categories, Materials (with low-stock threshold), Purchase orders & Purchases (receipt, qty, rate, total).

- Stock ledger: every purchase = IN, every usage/transfer = OUT; current stock = sum of movements.

3.8 Construction & Contractors

- Phases (foundation, structure, finishing…) per project, Tasks per phase with progress %.
- Contractors: contract value, scope, payment schedule, **Contractor Payments** (advance / running bill / final).
- Construction progress % derived from tasks; phase actual cost derived from real transactions (see §4).

3.9 Finance module (core of the Finance App)

Screen	Purpose
Accounts	Bank / cash / mobile-wallet accounts with balances; every transaction moves one account
Expenses	Company & project expenses with category, method, status, attachment
Project Costs / Budgets	Budget per project and per phase; compare against live actual cost
Cost Allocations	Owner funding goals (§3.6)
Reports → Cash Flow	Money in vs out by period
Reports → Categories	Spend by expense category
Reports → Contractor Payments	Paid / due per contractor
Reports → Owner Contributions	Payable, paid, outstanding, fines per owner & goal

Payment methods: Cash, Bank Transfer, Cheque, bKash, Nagad, Rocket, Card, Other.
Transaction statuses: draft, pending, submitted, verified, completed, cancelled, rejected.
Default expense categories (free text allowed): Construction, Labour, Contractor, Material, Equipment, Machinery, Transport, Site Work, Utilities, Office, Staff, Travel, Professional, Legal/Government, Miscellaneous.
Four finance families are kept **separate** (never one generic "transactions" table): Project (budget, expense), Customer (payment, invoice, receivable), Shareholder (contribution, payment), Company (income, expense, adjustment).

3.10 Documents, Approvals, Alerts, Reminders, Notifications

- Documents: attached to Company / Project / Unit / Customer / Shareholder / Landowner / Vendor / Transaction, with visibility rules (what portal users may see).
- Approvals: workflow for sensitive records; Alerts: budget overrun, overdue payments, low stock; Reminders & Notifications: due dates, follow-ups.

3.11 Portals (read-only views for outsiders)

- **Customer portal:** own units, installment schedule, payments, documents, dues.
- **Shareholder portal:** own holdings, contributions, outstanding, fines.
- **Landowner portal:** own agreements and allocated units.

3.12 Admin utilities

- Dashboard, Analytics, Activity feed, Audit log, Users & Roles, Company settings, Backup/export, Help.

4. Calculation rules (the “hisab-nikash”) — MUST follow exactly

> **Golden rule:** compute everything at *read time* from source transactions. Never store a total or status that can go stale. Always filter by status, and use adjusted (effective) values — never raw sums.

#	Quantity	Formula
1	Counts toward actual cost	Include a transaction only if status is `verified`, `completed` or has no status. Exclude draft, pending, submitted, cancelled, rejected.
2	Project / phase actual cost	Σ(Purchases) + Σ(Project Expenses) + Σ(Contractor Payments), each filtered by rule 1.
3	Budget variance	Budgeted − Actual(live). Warn at 80% used, critical at 100% (per-row threshold overrides allowed). A manually typed “actual” field must be labelled “Actual (Manual)” — never show it as the real actual.
4	Effective payable (owner)	payableAmount + Σ(adjustments). Original never mutated.
5	Outstanding	max(0, effectivePayable − paidAmount)
6	Overdue	outstanding > 0 AND dueDate < today
7	Fine	If overdue: flat fine = fineValue; percent fine = outstanding × fineValue ÷ 100; else 0.
8	Total due	outstanding + fine
9	Contribution status	outstanding ≤ 0 → **paid**; else overdue → **overdue**; else paid > 0 → **partially-paid**; else **pending**.
10	Installment obligation status	outstanding ≤ 0 → paid; due date passed → overdue; some paid → partially-paid; due within 7 days → due; else upcoming.
11	Total Payable (aggregates)	Σ *effectivePayable* (not raw payable), so Payable − Paid always equals Outstanding even with waivers.
12	Goal collection %	Σ paid ÷ Σ effective payable × 100.
13	Goal status	cancelled → archived → draft (admin decisions win) → completed (100% AND admin marked complete) → fully-funded (100%) → partially-funded (>0%) → upcoming (start date in future) → active.
14	Sales totals	Exclude cancelled / non-final bookings and sales (same status-filter idea as rule 1).
15	Customer receivable	Contract value (after discounts) − Σ valid payments; overdue = unpaid installments past due date.
16	Stock on hand	Σ IN − Σ OUT from the stock ledger; alert when below material threshold.
17	Cash flow	Σ money-in (customer payments, owner contributions, other income) − Σ money-out (counted expenses, contractor payments, purchases) per period.

Self-check: if the same real-world number appears on two screens and they differ, that is a bug — fix it, don’t ignore it.

5. Typical end-to-end flows

1. **Sale:** Lead → Customer → Booking (booking money) → Contract → Installment schedule → Payments → Receipt → Receivable reaches 0 → Handover document.
2. **Construction funding:** Create Cost Allocation → generate owner contributions → owners pay → system computes outstanding/fine each day → reminders → adjustments if waived → goal fully-funded → completed.
3. **Spending:** Budget set → Purchase order / Purchase or Expense or Contractor Payment (draft → submitted → verified) → counted in actual cost only when verified/completed → budget alert at 80% / 100%.
4. **Month end:** Cash-flow report, category report, contractor-payment report, owner-contribution report, receivables list; export for accountant.

6. Data model summary

Domain	Entities
Identity	User, Credential, Role
Property	Project, Building, Floor, Unit, Parking, Property
CRM / Customer	Lead, LeadActivity, Customer
Ownership	Shareholder, Shareholding, Landowner, Agreement, LandownerAllocation
Sales	Booking, Sale, Contract, Installment, CustomerPayment, Invoice, Receivable
Finance	FinanceAccount, ProjectBudget, ProjectExpense, CompanyIncome/Expense/Adjustment, CostAllocation, OwnerContribution, ContributionAdjustment, GoalInstallment, OwnerInstallmentObligation
Procurement	Vendor, MaterialCategory, Material, PurchaseOrder, Purchase, StockMovement
Construction	ConstructionPhase, ConstructionTask, Contractor, ContractorPayment
System	Document, Notification, Reminder, Alert, Approval, AuditLog, CompanySettings

Each entity stores money as an amount (+ currency BDT) and carries created/updated by and timestamps.

7. What the Finance App should do first

1. Implement §4 formulas as shared functions (one place), then build screens on top.
2. Build in this order: Accounts → Expenses/Project costs → Customer payments & installments → Owner contributions → Reports → Alerts.
3. Keep roles, audit log and status workflow from §2 and §4 rule 1.